

Hawler medical university College of Medicine



MEDICAL TERMINOLOGY

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OBJECTIVES

- You will be able to gain a better understanding of the most common suffixes and prefixes used in medicine and surgical procedures
- better understanding of the medical terms in association with different anatomical positions

RETRO = Behind. Examples:

Retrosternal pain is pain behind the sternum.

Retroorbital swelling can bulge the eyes forward.

Retrograde flow is a backward flow as opposed to normal forward flow.

Retrograde catheterization of urinary tract to discover point and cause of obstruction.

- ***INTRA*** = Within. Examples:
- **Intradural swelling** is one within the dura in the spinal canal.
- **Intravenous injections** deliver drugs in the veins.
- **Intramuscular injections** deliver drugs in the muscle.
- **Intraocular** pressure is the pressure inside the eyeball.
- **Intracranial** is within the skull.

EXTRA = Outside. Examples:

Extracellular space is the space outside the cells.

Extradural swelling is one outside the dura in the spinal canal.

Extracranial is outside the cranium or skull.

Extraordinary is going beyond what is usual, remarkably exceptional.



- ***ANTE*** = Before. Examples:
- **Antenatal care** refers to the care for the pregnant and fetus before birth.
- **Antemortem**: Before death.
- **Antecedents**: Ancestors, parents.
- **Antepartum hemorrhage** is vaginal bleeding before birth of the child.

POST = After. Examples:

Postnatal care: Care to the mother and neonate after birth.

Postmortem examination is the dissection and examination after death to discover the cause of death.

Postnasal drip is the flow of mucous secretion from the posterior part of the nasal cavity onto the wall of the pharynx.

Postpartum hemorrhage is vaginal bleeding after labor.



- ***PRE*** = Before, earlier than, in front of.
Examples:
- **Premedical studies** are those that prepare you to study medicine.
- **Premalignant conditions** are those that occur before and predispose to **malignancy**.
- **Presacral swelling** is one situated before or anterior to the sacrum.

CRYPTO = Hidden. Examples:

Cryptorchidism is undescended testis.

Cryptogenic cirrhosis is cirrhosis of unknown cause.

META = Change. Examples:

Metaplasia is a change from one type of epithelium into another.

In smokers the delicate pseudostratified ciliated columnar epithelium of the bronchial mucosa undergoes metaplasia into squamous type epithelium.

Metachromasia is undergoing a color change in certain reactions. The toluidine “blue” dye stains the granules of mast cells purple.

Metabolism is the sum of the chemical changes and reactions in living cells by which energy is provided and new material is assimilated.

ECTO = On the outside. Examples:

Ectoderm is the outer embryonic layer.

Ectocyst is the outer layer of a cyst.

Ectopic: away from the normal anatomic location. An ectopic **thymic tissue** may be found in the neck.

- ***EXO*** = Outside, away. Examples:
- **Exotoxin** is a toxic bacterial product secreted outside the cell.
- **Exophthalmus** is the outward bulge of the eyes in primary hyperthyroidism (Grave's disease).
- **Exocrine glands** deliver their secretions into ducts as in salivary glands.

ENDO = Inside, within.

Endocardium is the inner layer of the heart.

Endotoxin is a toxic bacterial product inside the cell that is liberated when the bacterium dies.

- **Endocrine glands** pour their secretions directly into the blood like the thyroid, adrenals, pituitary, etc.
- The pancreas has both endocrine portion (The islets of Langerhan secrete hormones like insulin) and an exocrine portion consisting of glands and ducts secreting digestive enzymes.
- **Endoscope** is an instrument used to visualize hollow organs or cavities (e.g. cystoscope, gastroscope, colonoscope, laparoscope).
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NOCT = Night. Examples:

Nocturnal dyspnoea is difficulty in breathing occurring at night during sleep.

Nocturia is increased urination at night.

- ***EN*** = Inside, within. Examples:
- **Encephalitis** is inflammation within brain tissue.
- **Encyst** is to enclose within a cyst.
- **Encapsulated tumor** is one enclosed within a capsule.
- **Entrap** is to catch in a trap.
- **Enuresis** is to urinate in bed.

ENTO = ENDO, within.

Entoderm = endoderm, the inner embryonic layer.

Entocornea is the inner corneal layer or Descemet's membrane.

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- ***EPI*** = Over, upon.
- **Epiglottis** (لسان المزمار) is located above and guards the Glottis (الحنجرة).
- **Epigastrium** (المنطقة الشرسوفية) is that part of the abdomen above the stomach.
- **Epidermis** is the outer keratinizing squamous epithelium covering layer of the skin

MESO = In the middle, intermediate.

Mesoderm is the middle germ layer in the embryo.

Mesothelium is epithelium derived from mesoderm e.g. the lining of pleural and peritoneal cavities.

Mesosalphinx, **mesovarium** and **mesocolon** are related to the peritoneal folds of the fallopian tube, the ovary and colon respectively.

- ***HETERO*** = Different. Examples:
- **Heterogeneous**: consisting of different constituents, mixed.
- **Heterograft**: An organ or tissue taken from a donor and grafted into a recipient of another species.
- **Heterotopia: Ectopic**. Presence at abnormal anatomical site, e.g. presence of an island of gastric mucosa in upper esophageal mucosa.
- **Heterozygous**: Having two different genes at corresponding loci of homologous chromosomes.

HOMO = Same. Examples:

Homogenous: uniform texture or structure.

Homozygous: Identical genes at corresponding loci of homologous chromosomes.

Homologous: Having the same position, value or relative structure.

Homosexual: Having sexual desire to a member of the same sex.

- ***LEPTO*** = Thin. Examples:
- **Leptomeninges** are the pia-arachnoid thin meningeal covering of the brain and spinal cord.
- **Leptophonic**: Has thin voice.
- **Leptospirosis**: Infection with certain bacteria of a thin spiral shape.

PACHY = Thick. Examples:

Pachymeningitis is inflammation of the dura, which is the thick outer meningeal covering of the brain and spinal cord.

Pachynoychia: Thick nails.

Pachydermatosis are skin diseases characterized by thickening of skin.

- ***MACRO*** = Large. Examoles:
- **Macrophage**: A large phagocytic cell derived from blood monocytes.
- **Macrocytosis** of erythrocytes: Enlarged size of red blood cells seen in megaloblastic anemia due to vitamin B12 or folic acid deficiency.
- **Macroscopic**: Large enough to be observed by the naked eye.

MICRO = Small. Examples:

Microorganism: Very small living organisms not seen by the naked eye.

Microscope: An optical instrument consisting of a set of lenses to produce enlarged images of objects.

Microcephaly is a congenital abnormality characterized by a small head.

THANK YOU